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Asymmetric Relations Between Needs and Data Availability in Predictive Maintenance Systems

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Abstract

Maintenance decisions can remain consequential when feasible condition information is persistently weak. This paper characterizes Limited-Data Predictive Maintenance (LD-PdM) as a regime in which indirect evidence materially informs such choices. An applicability checklist and a five-step procedure organize the assessment around a specified decision, its feasible evidence, and the assumptions connecting that evidence to condition. A completed railway-station decision record illustrates the procedure using hypothetical exposure interpretations and condition probabilities. The two-door comparison evaluates early inspection together with the service response after a positive finding. Dominance filtering removes policies that are worse across the retained models; minimax regret and maximin incremental gain then express different selection objectives. The former limits worst opportunity loss, whereas the latter protects the minimum improvement over the existing schedule. Their decision regions show why disagreement about the best inspection need not justify retaining that schedule. The contribution connects established PdM suitability assessment, indirect inference, and decision analysis through an inspectable action rationale. The example demonstrates conditional policy comparisons; estimation of the proxy relationships and evaluation of operational usefulness remain separate tasks.

Keywords: predictive maintenance; limited data; partial observability; proxy information; model uncertainty; railway stations; inspection planning; decision support; minimax regret

1. Introduction

Predictive maintenance (PdM) connects evidence about an asset to decisions about its future operation, inspection, servicing, or replacement. Diagnosis and remaining useful life (RUL) prediction are important components of this process, but the operational purpose is to choose an appropriate action. Reviews describe the development of PdM, machine-learning methods, and their relationship to industrial information infrastructures [1–3]. The availability of informative measurements influences which parts of this workflow can be implemented. It does not, however, determine whether a maintenance decision has to be made.

Limited evidence is already a recognized problem in prognostics and health management (PHM). Li et al. [4] review limitations in monitoring quantity and quality, economic and operational barriers, and data-efficient learning. Their scope includes maintenance execution, multimodal records, physical knowledge, fleet information, and uncertainty. The question here is how to assess a maintenance proposal when feasible condition measurements leave important uncertainty and the proposed action depends on interpreting operational records, service history, or engineering assumptions.

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This regime is termed *Limited-Data Predictive Maintenance* (LD-PdM). The term describes the evidence available for a maintenance choice. Consider infrastructure devices whose maintenance affects service continuity, while the feasible information budget provides little dedicated condition monitoring for each device. Passenger flows, event logs, service records, documentation, and engineering knowledge may nevertheless inform hypotheses about usage and maintenance needs. The central asymmetry is that the consequence of a maintenance choice and the availability of informative evidence need not increase together. High decision consequences do not ensure informative observations; abundant records do not ensure relevance to the decision.

PdM suitability assessment already combines criticality with technical, economic, organizational, and maintenance-planning constraints. The procedure of Tiddens et al. [5] includes Yes/No/Maybe judgments and focused feasibility studies. A checklist and a distinction between consequence and monitoring feasibility therefore do not establish a new method. The organizing question here is narrower: for an already specified maintenance choice, which indirect inferential links support its action rationale, and what happens when their plausible interpretations disagree?

Established decision analysis provides the basis for comparing the resulting actions. Srinivasan and Parlikad [6] assess condition-monitoring value using a partially observable Markov decision process (POMDP), while Memarzadeh and Pozzi [7] connect information value to inspection, monitoring, and system-level scheduling. Bayesian networks offer tools for integrating uncertain evidence [8]. Dominance screening and alternative criteria for decisions under uncertainty are also established [9]. The present assessment makes their inputs and commitments explicit for a maintenance choice with weak condition information.

The paper supplies an applicability checklist (Table 1), a five-step assessment procedure (Section 4.1), and a completed two-door decision record (Table 5). The numerical comparison isolates how routing interpretation and inspection sensitivity affect two declared selection objectives. The contribution is a conceptual synthesis with a conditional demonstration: exposure interpretations and condition probabilities are stipulated, and operational usefulness is not evaluated. The inspection example represents one decision within a PdM workflow, rather than a complete prognostic system.

Section 2 defines the regime and its asymmetries. Section 3 describes evidence and belief formation, and Section 4 connects them to maintenance choices. Section 5 works through a hypothetical station example. Section 6 discusses overlap with existing approaches, limitations, and evaluation questions; Section 7 concludes.

2. The LD-PdM Regime and Its Asymmetries

2.1. Operational definition and boundary cases

LD-PdM is defined relative to an asset or population, a maintenance choice, and a planning horizon. It applies when the following conditions hold together:

(1) The maintenance choice has a consequential operational or economic outcome.
(2) A condition quantity relevant to that choice remains weakly informed by the feasible set of condition-sensitive channels because of persistent deployment, access, or acquisition constraints.
(3) Proxy, contextual, historical, or prior evidence makes a material contribution to the assessment of future maintenance needs.

Here, persistent means that the limitation is expected to remain relevant over the decision horizon. Classification is conditional on a declared feasible-information set: record accessible channels, acquisition resources, and the deadline. A hard access constraint and an untested judgment that monitoring is uneconomic are different justifications. Expanding

Table 1. Applicability checklist for a specified maintenance choice and horizon. Each row requires a yes/no/unresolved answer supported by the indicated justification.

Criterion	Assessment question	Required justification
Consequential choice	Do feasible actions differ in an outcome relevant to the stated objective?	Name the function, actions, horizon, consequence, and case-specific reason that the difference matters.
Persistently weak condition information	Do available channels leave a condition distinction unresolved at the resolution needed for this choice?	State the condition distinction, required resolution, accessible channels, residual uncertainty, and persistent constraint.
Material indirect evidence	Does an indirect source or assumption affect that condition assessment or action comparison?	Trace the affected quantity and show the change under removal or revision of the input. A change of the winning action is not required.

that set is a separate monitoring-investment decision and may change the classification. Weak direct information alone does not imply that a useful proxy exists.

Table 1 turns the criteria into questions requiring case-specific evidence. For each row, record *yes*, *no*, or *unresolved*, with the justification and its source. All three affirmative answers establish applicability under the stated assessment. A negative answer excludes the choice from this definition; an unresolved answer leaves classification open. Neither outcome establishes whether a particular inspection order is justified. That requires the separate action comparison in Section 4.

Three stipulated cases distinguish the outcomes. Clearing a directly observed obstruction fails the weak-information criterion for that response, so its classification is *no*. The early-inspection choice in Table 5 has three conditional *yes* answers: service consequences matter, the actionable condition is unobserved, and routing changes its assessed probability. Passenger counts with no justified link to device activity leave the third answer *unresolved*; record volume alone cannot close it. These examples demonstrate how the criteria branch, rather than establish assessor agreement or field usability. Few labelled samples with an informative condition channel can instead create a small-data learning problem. Weak condition information can also coexist with a robust action, so an unresolved state need not imply an unresolved choice.

The four asymmetries have different logical roles. Observability and acquisition constraints explain the defining mismatch between maintenance need and feasible condition information. Sparse or selective failure records complicate learning within that regime. Population scale creates opportunities for sharing evidence and coordinating work, together with additional dependencies. Rare failures and large populations are therefore not additional inclusion requirements. Low net information value is also not a measure of weak observability: information content, its effect on a choice, and acquisition cost must be assessed separately.

2.2. *Asymmetry of observability*

The operational importance of a variable does not guarantee an informative measurement channel. A planner may need to anticipate loss of service while observing only traffic, location, occasional alarms, or service interventions. Such records can constrain hypotheses about device usage without identifying condition. Figure 1 separates the available evidence from the model-dependent quantities required for decisions.

In the formulation of Tinga [10], usage- and load-based maintenance connects operating information to condition and maintenance intervals through physical failure models.

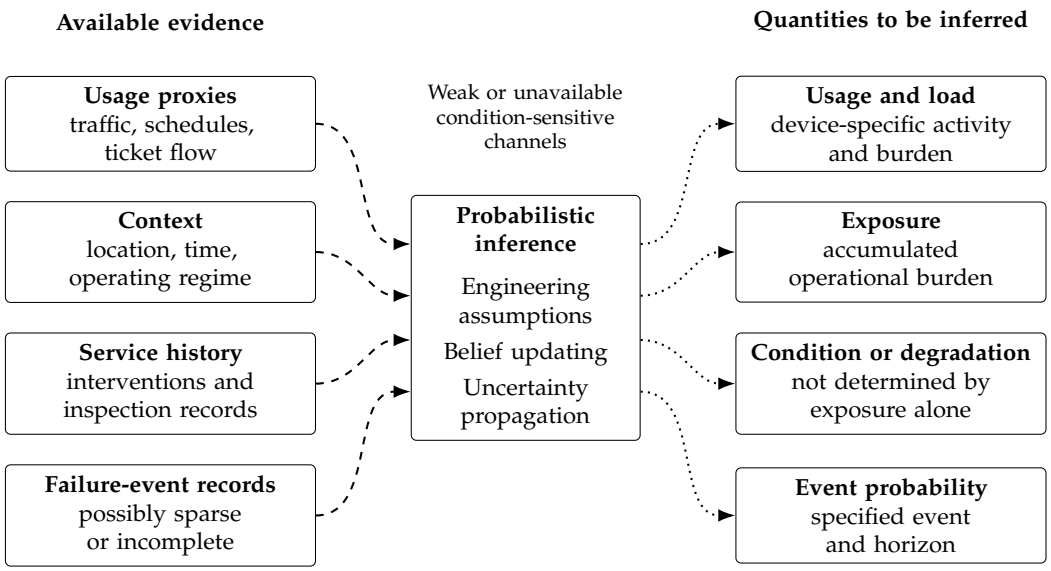


Figure 1. Conceptual separation between evidence, probabilistic reasoning, and maintenance quantities. Dashed arrows denote evidence inputs and dotted arrows inferential outputs. The common block leaves individual dependencies unspecified. This is an architecture of possible inference links; the numerical example stipulates exposure and condition probabilities and evaluates the final action comparison.

Soft sensors estimate process quantities from other measurements using model-driven or data-driven relationships, with model validation and changing operating conditions as practical concerns [11]. In the station setting, even device usage may be indirectly known. The assessment must therefore identify both the target of each estimate and the evidence supporting the next link to maintenance need.

2.3. Asymmetry of failure evidence

Operation and maintenance generate selective histories. Replacement before failure may censor a lifetime, periodic inspection may locate an event only within an interval, and service changes the physical state. Missing work orders are a separate source of uncertainty. Cooke and Bedford [12] discuss the interpretation of reliability databases when preventive maintenance and failure are dependent. Consequently, the absence of a recorded failure cannot automatically be treated as healthy-condition evidence. An empirical model must distinguish survival, intervention, and missing records and justify how observation and maintenance policies enter its likelihood. The present example stipulates condition probabilities and makes no estimate from such histories.

2.4. Asymmetry of information costs and decision consequences

The value of a maintenance decision and the cost of improving its evidence are different quantities. The latter includes acquisition and installation, integration with existing systems, commissioning and validation, connectivity and data processing, and the continuing operation and maintenance of the monitoring arrangement. The importance of deployment cost is also recognized in small-data PHM [4]. It motivates a conditional applicability statement: LD-PdM is relevant where better dedicated information cannot presently be obtained within the operator's feasible resources and constraints.

Table 2. Quantities distinguished in the conceptual framework. The final column identifies an inference limit, rather than a missing numerical parameter to be supplied in this paper.

Quantity	Meaning	Interpretive limit
Evidence	Proxy observations, context, service records, documentation	Record volume does not establish condition information
Usage	Activity during a specified interval, such as opening cycles	Passenger traffic is not a device-cycle counter
Exposure	Accumulated mechanism-relevant burden over a stated history	Equal cumulative activity need not imply equal wear
Condition	Physical degradation or functional state	Similar exposure does not imply equal condition
Event probability	Probability of a specified event within a declared horizon	A service threshold and a physical failure are different events
Risk	Consequences evaluated together with event uncertainty	An event probability alone does not specify consequences
Priority and action	A choice given beliefs, consequences, and constraints	Priority is a decision output, not a physical state

This condition must be assessed for the particular asset function and decision. Unit purchase price does not determine criticality, and an elevator or door may have substantial safety, accessibility, or service consequences. Additional monitoring can therefore be justified for a consequential function even in a population of ordinary devices. The station example supplies no asset-class estimate of sensing profitability. Condition-monitoring value depends on what the information changes in the maintenance policy [6]; existing operational data also have integration and interpretation costs.

2.5. Asymmetry of scale

Uncertain individual states can coexist with useful population-level questions: where exposure concentrates, which groups merit inspection, or how limited attention should be distributed. Hierarchical modelling can share information across appropriately related units; Bayesian IoT failure modelling provides one example [13]. Risk-based and opportunistic maintenance studies illustrate the importance of system consequences and coordinated interventions [14,15]. These are established modelling opportunities, not a guarantee that a larger population makes weak evidence sufficient.

Partial pooling requires a defensible relationship between units, including their mechanisms, environments, and maintenance histories. A hierarchical model should allow between-unit variation where these differ; complete pooling would impose a stronger common-parameter assumption. Shared routing errors or a common misspecified proxy relationship do not disappear when more assets are included. Heterogeneous asset classes may require different models. Furthermore, estimating total burden and selecting the particular asset requiring action are different tasks: identical evidence and priors do not justify a data-derived individual ordering. The operator must therefore distinguish estimating population burden from choosing an individual asset for inspection.

3. Evidence and Belief Formation

3.1. Separating observations from maintenance quantities

A proxy is informative only through a relationship with a specified target. Passenger counts might constrain use of an access route; a device-use model might translate that flow into operating cycles; cycles might inform cumulative exposure. None of these links, by itself, measures damage or establishes a failure probability. Age, maintenance quality, environment, design, and latent defects may alter the relationship between exposure and condition. Table 2 distinguishes the objects needed to describe that chain.

The evidence may combine operational aggregates, contextual variables, inspection and intervention histories, and engineering documentation. Their roles should remain identifiable. A timetable constrains possible demand; a service record describes an intervention; documentation supplies a mechanism or service requirement. Treating these as interchangeable features obscures observation assumptions. Heating, ventilation, and air-conditioning (HVAC) fault detection using ambient measurements and electrical-current inputs provides relevant prior art for contextual inference [16]; maintenance decision systems also integrate heterogeneous information [17]. Their measurement requirements must be compared with the station's feasible channels.

3.2. Representing the unresolved quantities

The assessment should identify which quantities are observed, inferred, or supplied by engineering judgment. For a door, usage can be the number of opening cycles in a recording interval, whereas exposure accumulates relevant activity over a stated history. A wear model may weight cycles differently according to load or environment. The history over which exposure is accumulated must be distinguished from the future calendar-time window over which the maintenance choice is evaluated. No universal conversion from passenger totals to either quantity is assumed here.

A probabilistic representation can connect these quantities while preserving uncertainty about their relationships. Bayesian updating supplies established tools for learning from observations [18]; Bayesian networks and expert systems can represent dependencies and judgment-based assumptions [8,19]. The representation should be as specific as the decision requires. Section 5.3 uses a probability of an actionable condition at inspection, together with detection and consequence assumptions. It does not require a full degradation trajectory or a physical-failure probability. Either extension would need an additional transition model and an explicit event definition.

3.3. Priors, dependence, and limits of identification

Engineering knowledge may constrain plausible routing, device-use mechanisms, deterioration sensitivities, or service conditions. Such information belongs in an explicit assumption register with its source and scope. Documentation, expert judgment, and previously observed behaviour need not have equal evidential strength. Multiple records derived from the same original traffic estimate are also not independent confirmations. A model should preserve their dependence rather than treating duplication as increased support.

When observations weakly constrain a parameter, its posterior can remain strongly dependent on the prior or chosen model class. Honest uncertainty representation therefore includes sensitivity to plausible alternative assumptions and identification of links that the available evidence cannot resolve. In LD-PdM, this can mean stopping at an exposure comparison rather than extending the chain to a numerical failure probability. Inspection or a targeted measurement may then be more useful than a more elaborate inference algorithm applied to the unchanged evidence.

3.4. Shared error and population comparisons

A simple exposure-error identity distinguishes population averaging from pairwise comparison. Let $\widehat{x}_i = x_i + b_s + \varepsilon_i$, with fixed true exposures, a shared zero-mean error b_s of variance σ_s^2, and mutually independent zero-mean asset errors of variance σ_ε^2, independent of b_s. For n assets,

$$\mathrm{Var}(\overline{\widehat{x}} - \overline{x}) = \sigma_s^2 + \frac{\sigma_\varepsilon^2}{n}, \qquad \widehat{x}_i - \widehat{x}_j = (x_i - x_j) + (\varepsilon_i - \varepsilon_j). \tag{1}$$

The shared component remains in the mean and cancels in the exposure difference. Here n counts assets, not repeated records. Conditional on a known fixed bias its variance is zero; uncertainty about an unknown bias may still be represented probabilistically. Route-specific errors need not cancel. More importantly, cancellation in exposure differences need not survive the condition-to-action mapping, as Section 5.4 shows. This identity is an exposure-level caution, not a fleet decision model.

4. Maintenance Choices under Weak Information

4.1. A procedure for assessing a maintenance proposal

The procedure takes an asset function, a maintenance choice, available records, and operator constraints as inputs. It produces a documented action rationale or identifies the missing evidence or decision commitment that prevents selection:

1. **Specify the decision.** State the mechanism, future planning window, consequences, feasible complete policies, resource limit, and mandatory requirements. Record the existing feasible schedule as the reference choice.
2. **Assess applicability.** Complete Table 1. Identify feasible condition-sensitive channels and the quantity they leave unresolved. Explain why the constraint persists over the window and how indirect evidence affects the assessment.
3. **Trace the evidence.** Link each record to the quantity it informs. Separate observed, inferred, and stipulated inputs. Declare the retained model or parameter set and its justification; exclude interpretations contradicted by established evidence. Record shared sources and observations that could change the set.
4. **Compare feasible policies.** Include the action following each inspection outcome. Evaluate gains relative to the schedule under every retained model. Remove dominated policies and declare the objective used among the survivors, including whether a loss against the schedule in any model is allowed.
5. **Record and act.** State the criterion's selection or tie set, assumptions, and reconsideration trigger. If a unique action is needed, document the tie-break or missing commitment. Do not restore a dominated policy merely because model-specific winners disagree. New mandatory requirements change the feasible set and require reassessment.

Classification determines which evidence links require scrutiny; it does not prescribe a new maintenance algorithm. Three affirmative answers require the indirect links and their decision sensitivity to be documented in steps 3–4. A negative answer redirects the choice to the applicable existing process. An unresolved answer identifies a missing justification; only justified inputs may enter the action comparison. Improvement over the baseline, absence of dominance, and optimality under a declared criterion remain separate judgments.

4.2. Comparing inspection policies

An inspection policy specifies both the initial inspection and the action after each possible observation. Let π denote such a policy, Π the feasible set, E the evidence at the planning review, and m an assumed model. Over the common future window, write

$$R_m(\pi) = \mathbb{E}_m^{\pi}[L \mid E], \qquad \pi_m^{\star} \in \arg\min_{\pi \in \Pi} R_m(\pi). \tag{2}$$

Here L includes inspection, intervention, disruption, and any terminal consequence within the window. The superscript indexes prospective evolution and observations under the chosen policy. Beliefs formed from E must account for the historical observation and intervention process when it is informative; prospective policy choice does not make that process ignorable. The finite example below stipulates its condition probabilities instead of

estimating this likelihood. POMDP and information-value maintenance analyses supply fuller sequential formulations [6,7].

An inspection has value when a possible finding changes a feasible subsequent action enough to justify the inspection burden. Greater uncertainty alone establishes neither this value nor a reason to intervene. A highly uncertain quantity may be irrelevant to the available actions, while a relatively well-understood defect can warrant prompt service. A counting exercise and a physical inspection also answer different questions: the former may resolve route allocation, while the latter may reveal a condition requiring action. Valuing either requires its observation model, cost, and possible effect on the subsequent decision.

4.3. Resources and robustness

Across assets, comparison requires a common objective and resource constraint. Exposure describes burden; inspection value additionally depends on detection and the feasible response. Maintenance studies address system consequences, coordinated interventions, and practical resource constraints [14,15,20,21]. In a station, taking one asset out of service may change another's use or remove an accessible route, requiring a joint consequence comparison.

For nonempty finite policy and model sets, let $\mathcal{M}$ denote the declared plausible models and $G_\pi(m) = R_m(\pi_0) - R_m(\pi)$ the expected gain over the reference schedule, with $G_{\pi_0} = 0$. First retain only the undominated set Π_{ND}: remove π if another feasible policy has at least as much gain in every model and strictly more in at least one. Feasibility concerns constraints; this dominance screen concerns comparative consequences [9].

Two objectives then make different commitments:

$$\begin{aligned} \Pi_{\mathrm{MG}} &= \arg\max_{\pi\in\Pi_{\mathrm{ND}}} \min_{m\in\mathcal{M}} G_\pi(m), \\ \Pi_{\mathrm{MR}} &= \arg\min_{\pi\in\Pi_{\mathrm{ND}}} \max_{m\in\mathcal{M}} \left[\max_{\rho\in\Pi} G_\rho(m) - G_\pi(m)\right]. \end{aligned} \tag{3}$$

Maximin incremental gain (MG) protects the smallest improvement over the schedule. Minimax regret (MR) limits the largest opportunity loss relative to the best policy within each model. The example uses MR as its declared main objective and MG as a comparator, among three deterministic policies. Neither criterion requires probabilities over the models. Exact ties remain sets; a required unique action needs a documented tie-break. With finitely many policies and continuous gains, the extrema are also attained over compact parameter sets, as in the later rectangle stress test.

MR can accept a loss against the schedule in one model to reduce regret in another. If nonnegative gain in every model is mandatory, impose that additional constraint before selection; it is not imposed here. MG is not generally minimax absolute loss because $R_m(\pi_0)$ may vary across models. Those unknown baseline terms cancel in within-model dominance and regret. Recommendations remain conditional on the retained model set, which an omitted interpretation can invalidate.

5. Worked Conceptual Example: A Railway Station

5.1. Setting and assumed evidence

Consider a hypothetical station with a concourse, platform access routes, automatic doors, elevators, and escalators. The example assumes that accessible condition-sensitive channels are insufficient for the specified choice. An actual assessment would first inventory controller counters, alarms, service records, and OEM or contractor data, distinguishing missing measurements from unavailable access. A usable cycle counter could remove usage uncertainty while leaving condition uncertain. No asset-class claim about absent

instrumentation or sensing profitability follows from the scenario. Consequences and mandatory requirements depend on the particular function and its accessible alternatives.

Suppose the planner has zone-level passenger counts or ticket-flow summaries, timetables and platform assignments, a station layout, device documentation, and dated inspection and service records. At the present review, there is capacity for at most one additional early inspection before the next scheduled review. A predefined service response after a positive finding is also assumed feasible. Existing mandatory work and the established service schedule remain in force. The future window is a common interval of calendar time; historical exposure is accumulated over a common past interval ending at the present review.

Table 5 completes the assessment for one numerical case. Its sources are the declared scenario premises and equations. No passenger records or channel-access audit from an operating station are supplied. The example fixes conditional inputs to examine their decision implications; the preceding passenger-to-exposure chain identifies what an application would need to establish.

5.2. From passenger context to conditional beliefs

Passenger information first informs possible device use through an assumed routing relationship. Counts at an entrance do not identify the route taken by every passenger or the device selected. Timetables and platform assignments can constrain plausible routes, while closures and alternative access paths can change them. Routing uncertainty must remain attached to any resulting device-level usage belief. Ticket-flow and passenger-count summaries may also share an upstream source; their agreement is then weaker corroboration than agreement between independent observations.

Station topology gives this interpretation a system-level structure. A graph of entrances, circulation zones, platform connections, and accessible alternatives can encode which devices could serve a passenger stream without claiming to reconstruct individual trajectories. An outage or planned intervention may redirect that stream and change use of neighbouring assets. Device exposure is consequently conditional on route availability as well as demand. If the planner considers taking a device out of service, the resulting redistribution belongs to the consequences of that choice; the pre-intervention exposure map cannot simply be assumed to remain valid.

The next link is device-specific. For an automatic door, passage demand can inform opening cycles only through assumptions about grouped passage, reopening, controller behaviour, and timing. One opening may serve several passengers, so equal traffic totals with different clustering need not imply equal activity. Elevator trips and escalator run time require different mappings. A common passenger indicator therefore supplies no common wear score across these mechanisms.

Usage may then inform a model of accumulated exposure over a stated interval, conditional on the mechanism being considered. Exposure is an intermediate description of operational burden, not an observation of damage. Connecting it to physical condition requires further assumptions about initial state, maintenance effects, environment, and the applicable degradation mechanism. A service record can modify those beliefs without establishing that the device was restored to an as-new state. Absence of a recorded fault is also compatible with incomplete reporting or earlier preventive intervention. Table 3 identifies where additional evidence could challenge the chain.

An event probability requires an explicit event model: reaching a service condition, detecting a defect, and losing function before the next review are different events. The related door-and-elevator study develops Bayesian proxy-flow inference, age/cycle service-condition thresholds, and maintenance scheduling [22]. Its term “Low-Data” and this

Table 3. Assumption register for the hypothetical station. None of these relationships is estimated or validated in the present paper.

Inferential link	Assumption required	Evidence that could challenge it
Station flow to device use	Coverage, routing, and device-choice assumptions apply during the interval represented by the observations.	Local passage observations, closures, or changed platform use.
Device use to exposure	The selected mechanism and operating mode explain the usage quantity being accumulated.	Cycle/run-time records, controller settings, or changed operating practice.
Exposure to condition	Relevant initial-state, degradation, and maintenance-effect assumptions are credible for this asset.	Inspection findings, component history, or signs inconsistent with the assumed mechanism.
Condition to future event	A specified event, horizon, and applicable conditional model are available.	Dated event/inspection records and checks of predictive calibration.
Beliefs to action	Condition probability, detection, net response value, inspection burden, and constraints justify the comparison.	Inspection findings, response outcomes, operator review, and access dependencies.

paper's "Limited-Data" refer to related motivations; the latter is retained for the present decision-regime characterization. The present contribution is the assessment record and explicit comparison of action criteria, rather than a new routing estimator. No technical equations, numerical results, or figures are reproduced from the companion, and its service-condition thresholds are not treated as calibrated physical-failure probabilities here.

5.3. A symbolic two-door inspection comparison

Consider doors A and B on alternative concourse-to-platform routes. The reference policy π_0 retains the existing feasible schedule. Policy π_A uses the optional slot to inspect A early and performs the predefined service response after a positive finding; after a negative finding it retains the schedule. Policy π_B does the same for B. All policies have the same available capacity and service obligations, although π_0 leaves the optional slot unused. No mandatory requirement determines the order in this example. If it does, that requirement takes precedence over the following comparison.

An actionable condition is defined by a predeclared component or service-limit criterion for which the specified response is applicable. The criterion is assumed available in this hypothetical comparison; no physical threshold is calibrated here. For door $i \in \{A, B\}$ and model m, let $q_i(m)$ be its probability of being present at early inspection and $d_i(m)$ the inspection sensitivity, conditional on its presence. Let $k_i(m) \geq 0$ be the incremental inspection burden incurred regardless of the finding. Let $v_i(m)$ be the conditional expected net loss avoided by the response given a true positive, relative to π_0. It includes effectiveness, service effort, and disruption within the window, excluding the inspection burden. The losses share one scale.

Assume for this bounded comparison that false positive results do not occur. Inspection without a response is assumed to leave the physical state and schedule unchanged. A true positive has probability $q_i d_i$ and conditional net benefit v_i; the remaining outcomes retain the reference policy. Conditioning on these outcomes in Equation 2 gives

$$G_i(m) = q_i(m)d_i(m)v_i(m) - k_i(m), \qquad R_m(\pi_i) = R_m(\pi_0) - G_i(m). \tag{4}$$

Thus G_i is the expected improvement from a complete inspection-and-response policy. With condition C_i, detection D_i, and net avoided loss V_i, the benefit factor is

$P(C_i)P(D_i \mid C_i)\mathbb{E}[V_i \mid C_i, D_i]$. Detection and benefit can depend on severity; their independence is unnecessary because v_i is conditional on a true positive. If false positives occur, subtract $(1 - q_i)f_i h_i$, where f_i is their conditional probability and h_i the conditional extra loss of the erroneous response. Cross-door effects must be negligible or included in the conditional losses; independent door states are not required. Negative findings leave undetected condition in the baseline consequences.

Suppose the routing, grouping, and controller assumptions support greater historical cycle exposure for A. Under comparable histories and an assumed increasing exposure–condition relationship, this may support $q_A > q_B$. That additional relationship is not learned from the traffic proxy. For uncertain exposure, ordering mean exposure alone need not order q_i under a nonlinear relationship; the condition probability must account for the exposure distribution. With equal inspection burdens and positive $d_i v_i$, Equation 4 yields the explicit reversal condition

$$G_B > G_A \iff \frac{d_B v_B}{d_A v_A} > \frac{q_A}{q_B}, \qquad q_A > q_B > 0, \quad d_A v_A > 0. \tag{5}$$

Equal positive detection probabilities and response values preserve A's order under the equal-burden assumption. A lower-exposure door can warrant earlier inspection when detection or response benefit is sufficiently greater. The schedule must also remain in the comparison: under a single model, it dominates an inspection with negative gain.

Two evidence states must be distinguished. If dated records establish that the original allocation assigned A passengers who used a third passage during the assessed period, the contradicted routing model is excluded. Correcting that historical interpretation revises the exposure estimate; a later rerouting could not erase accumulated cycles. If the records merely leave both interpretations plausible, retain both in $\mathcal{M}$ and apply Equation 3. The criterion resolves their disagreement without requiring a common strict winner. A counting exercise would require its own observation likelihood, cost, timing, and effect on selection before an information value could be calculated.

The example also clarifies the treatment of maintenance records. It conditions on assumed current condition probabilities and does not estimate them from censored lifetimes. In an empirical application, earlier replacement could censor a lifetime, selective inspection could change which conditions become recorded, and service could change the state itself. Those processes would belong in estimation of q_i and the response consequences. Treating missing or selectively collected records as healthy observations would not justify the inputs to Equation 4.

5.4. Numerical illustration and continuous decision boundaries

Write dimensionless accumulated exposure as $x_i = X_i / X_{\text{ref}}$, where X_i is historical cycle exposure and $X_{\text{ref}} > 0$ is a common reference exposure. Two interpretations of the routing records assign $(x_A, x_B) = (1, 0.5)$ under m_1 and $(0.35, 0.5)$ under m_2. The latter represents passengers diverted from A to a third open passage during the assessed period. It does not transfer them automatically to B. The illustrative condition model is $q_i = 0.1 + \beta x_i$, with $\beta \in [0, 0.5]$. Inspection sensitivity is $d_B = 0.9$ and $d_A \in [0.25, 0.9]$; the common net response value is $v_i = 10$ and inspection burden $k_i = 1$, in arbitrary loss units. False positives are excluded as in the compact model. These values are declared assumptions, not station measurements, elicited uncertainty bounds, or estimated inspection performance. Every probability remains within $[0, 1]$ over the stated ranges.

Table 4. Exact hypothetical inspection comparisons under the two routing models. Gains use arbitrary loss units; 0 denotes the existing schedule and braces denote ties. Both criteria operate on the undominated deterministic policies. Minimax regret can select a policy with negative gain under one retained model; maximin refers to incremental gain relative to the model-specific baseline.

β	d_A	$G_A(m_1)$	$G_A(m_2)$	G_B	Minimax regret	Maximin gain
0	0.9	-0.1	-0.1	-0.1	0	0
0.2	0.9	1.7	0.53	0.8	A	B
0.2	0.4	0.2	-0.32	0.8	B	B
0.2	0.6	0.8	0.02	0.8	B	B
$\frac{1}{45}$	0.9	0.1	-0.03	0	A	$\{0, B\}$
0.5	0.9	4.4	1.475	2.15	A	B

Substitution into Equation 4 gives

$$G_A(m_1) = d_A(1 + 10\beta) - 1, \quad G_A(m_2) = d_A(1 + 3.5\beta) - 1,$$
$$G_B = 4.5\beta - 0.1, \qquad c(\beta) = \frac{0.9(1 + 5\beta)}{1 + 10\beta}. \tag{6}$$

Here $c(\beta)$ is the A/B tie boundary under m_1: A has greater gain above it and B below it. B improves on the baseline exactly when $\beta > 1/45$. A's baseline boundary under m_1 is $d_A = 1/(1 + 10\beta)$. Both comparisons matter near $\beta = 0$, where a higher-ranked early inspection can still have negative gain.

At each fixed (β, d_A), take $\mathcal{M} = \{m_1, m_2\}$, with no probabilities over the interpretations. The condition and detection probabilities within each model coexist with this model ambiguity. Under m_2, B has greater gain than A whenever $\beta > 0$, and also at $\beta = 0$ if $d_A < 0.9$. A common strict winner therefore need not exist even when both inspections improve on the schedule. Dominance filtering and Equation 3 address that distinction.

Table 4 compares the criteria using exact values. At $\beta = 0.2, d_A = 0.9$, A's gains are 1.7 and 0.53, whereas B's gain is 0.8 under either route. Both dominate the schedule. A's maximum regret is $0.8 - 0.53 = 0.27$; B's is $1.7 - 0.8 = 0.9$. MR selects A, while MG selects B because $0.8 > 0.53$. Retaining the schedule solely because neither inspection is strictly best under both models would discard a guaranteed improvement. Reducing d_A to 0.4 makes B preferable despite A's greater exposure under m_1, and both criteria select B. If dated evidence instead establishes m_2 and excludes m_1, both criteria select B in these cases under the reduced model set.

The criteria also expose a substantive tradeoff. At $\beta = 1/45, d_A = 0.9$, A has gains 0.10 and -0.03, whereas B and the schedule both have zero gain. MR chooses A, accepting the model-specific loss to reduce worst regret; MG ties B with the schedule. Thus removal of dominated policies is not a guarantee of nonnegative gain in every model. The selection objective and any additional improvement requirement belong in the action record.

Across the declared rectangle, MG selects B for $\beta > 1/45$, the schedule for $\beta < 1/45$, and both at equality. MR has the boundaries

$$r(\beta) = \frac{1.8 + 9\beta}{2 + 13.5\beta} \quad (\beta > 1/45), \qquad t(\beta) = \frac{2}{2 + 13.5\beta} \quad (\beta < 1/45). \tag{7}$$

Above r, MR selects A, and below it B. Above t, it selects A, and below it the schedule; only the portion inside the rectangle applies. Each curve is a tie boundary. At $\beta = 1/45$, MR selects A above $d_A = 20/23$, ties the schedule with B below it, and ties all three at equality. Figure 2 shows these objective-dependent regions.

Treating both routes and the entire parameter rectangle as one uncertainty set is a different decision. Gains and gain differences are bilinear or affine, so their extrema occur

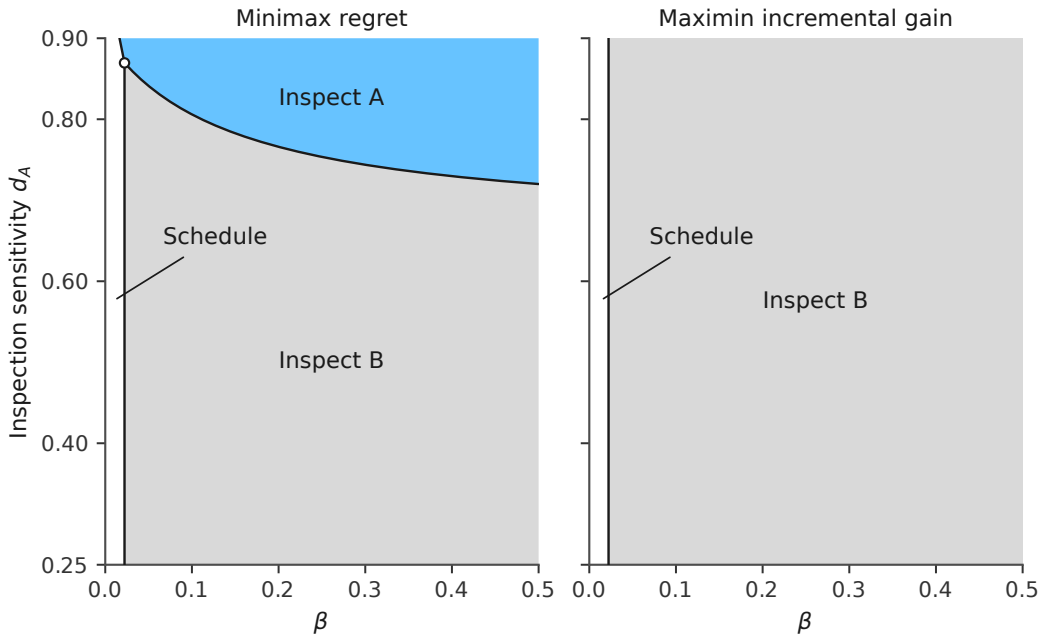


Figure 2. Policy selections after dominance filtering at each fixed (β, d_A), retaining both routing models. Left: minimax regret (MR). Right: maximin incremental gain (MG). Region labels identify A, B, or the existing schedule; dark boundaries denote ties and the open circle denotes the three-way tie. MR's narrow low-β region is governed by $t(\beta)$, with the three-way tie at $(1/45, 20/23)$. The axes form a hypothetical sensitivity domain, not a probability distribution or an empirical frequency of decisions.

at corners. A's gain then ranges from -0.75 to 4.4, and B's from -0.1 to 2.15. MG selects the schedule, whose guaranteed gain is zero. MR selects B: its maximum regret is 2.25, compared with 2.4625 for A and 4.4 for the schedule. This stress test does not turn the sensitivity rectangle into an elicited uncertainty set. The table, boundaries, and figure geometry are generated by a deterministic calculation from the declared input file, without fitted data or random sampling.

The shared-error caution also reaches the decision comparison. With the affine condition model, a common exposure shift s changes the gain difference by $\beta s(d_A v_A - d_B v_B)$. Thus cancellation in $x_A - x_B$ does not generally imply cancellation in $G_A - G_B$. Near the existing tie $\beta = 0.2, d_A = 0.6$ under m_1, small positive or negative shifts can change the preference while exposures remain positive. The shared-error identity is therefore insufficient to certify decision robustness.

5.5. Completed decision record and population limits

Table 5 assembles the procedure for the equal-sensitivity case. The main criterion requests inspection of A with its contingent response; the comparator identifies the different choice obtained by protecting minimum gain. The record makes both the evidence assumptions and the normative objective inspectable.

Extending the record to a population requires shared-source dependencies, defensible pooling relationships, coordinated feasible policies, and their joint consequences. Combining doors, elevators, and escalators because they share a zone supplies none of these assumptions. An intervention can also redistribute route use. The two-door calculation demonstrates a local choice within this system context; it does not implement fleet inference or coordinated station optimization.

Table 5. Completed conditional assessment for $\beta = 0.2, d_A = 0.9$. Sources are the declared premises and equations of the example, not station observations.

Step	Recorded assessment
1. Specify	Choose at most one additional early inspection before the next scheduled review. Feasible policies are π_0, π_A, π_B, including the predefined response after a positive finding. Existing mandatory work and the schedule remain in force.
2. Classify	Conditional yes/yes/yes: policies differ in assumed net loss; accessible channels leave the actionable condition unresolved; routing changes q_A. The channel constraint is a scenario premise, not an empirical audit.
3. Trace	Retain $\mathcal{M} = \{m_1, m_2\}$: neither interpretation has been excluded. Fix $\beta = 0.2$, $d_A = d_B = 0.9$, and the other inputs of Section 5.4. Exposures and the exposure–condition relation are stipulated. The rectangle is a separate sensitivity domain.
4. Compare	Gains under (m_1, m_2) are $\pi_0 : (0, 0)$, $\pi_A : (1.7, 0.53)$, $\pi_B : (0.8, 0.8)$. Both inspections dominate π_0. Maximum regrets for A/B are 0.27/0.90; minimum gains are 0.53/0.80.
5. Record	MR selects A and MG selects B. The declared main objective requests inspection of A with the predefined response. Reassess if dated route evidence excludes an interpretation, a condition/detection assumption changes, or a service constraint changes.

6. Discussion

6.1. Relationship to established approaches

Small-data PHM addresses economic and operational constraints, uncertainty, records, fleet information, and physical knowledge [4]. PdM suitability assessment already examines criticality and technical, economic, and organizational feasibility [5]. Table 6 places the present decision record alongside these approaches and the methods that can supply its inputs. The distinction concerns organizing purpose, not exclusive capabilities.

The specific output is a documented chain from the information-regime judgment to an action rationale, including model exclusions, dominance, and the selection objective. The two-door record exposes the conditional inputs needed to convert an exposure order into an inspection choice. Its expected-loss arithmetic and decision criteria are established tools. The worked record demonstrates internal coherence and inspectability; agreement between assessors, reduced omissions, and better operational decisions would require separate evaluation.

These perspectives can coexist in one application. A soft sensor or physical usage model may supply an input, and a POMDP may implement a sequential policy when its observation and transition models are justified. Arcieri et al. [23] incorporate inferred model uncertainty in railway maintenance planning through Bayesian decision analysis and POMDPs. Here, model probabilities are unspecified and the comparison is over a declared set. This narrower illustration does not replace posterior inference or establish a new solution to partial observability. Its inspection-choice focus also overlaps condition-based and risk-based planning; the retained LD-PdM term identifies the proposed information regime, not a claim that a prognostic model has been implemented.

Digital representations can retain asset identity, topology, histories, and assumptions [24–26]. Their role here is to preserve the provenance of a recommendation and the record that would trigger its revision. A full digital twin is not a prerequisite for applying the checklist or recording the finite policy comparison.

6.2. What a later evaluation should establish

Evaluation of an implementation should separate proxy interpretation, decision consequences, and usability of the assessment. Independent observations can test calibration

Table 6. Representative neighboring approaches and their role in the proposed assessment. The comparison concerns organizing questions and assumptions, not mutually exclusive method classes.

Approach	Question and required evidence	Role in the procedure
Small-data PHM [4]	How can PHM use limited quantity or quality of evidence, including records and physical knowledge?	Distinguish sample limitations from weak condition information before choosing learning methods.
PdM suitability assessment [5]	Which candidates merit PdM given criticality and technical, economic, and organizational feasibility?	Establish the feasible evidence set; then assess the indirect links supporting a specified maintenance choice.
Usage/load-based maintenance [10]	How do usage and loads affect condition and maintenance intervals under a physical model?	Check the usage-to-condition assumptions, including cases where usage itself is inferred.
Soft sensing [11]	Can other measurements and a validated relationship estimate an unavailable process quantity?	Identify the target of each proxy and the evidence for applying its relationship.
POMDP and information-value maintenance [6,7]	Which observation and maintenance policy is justified by condition uncertainty, costs, and consequences?	Compare complete inspection policies after specifying actions, observation quality, and constraints.

and transfer of proxy relationships across contexts. Complete policies can be compared through consequence-weighted missed events, unnecessary work, and information value under equal resource constraints. Assessor agreement and the completeness of decision records can test whether the procedure makes missing assumptions easier to identify. None of these evaluations is supplied by the conditional calculation.

Exact reconstruction of condition is unnecessary when all retained models support the same feasible action. Near a decision boundary, however, a small change in sensitivity or response value can change the choice. Validation should therefore target the uncertain distinctions relevant to that choice as well as the estimated quantities. A synthetic experiment with known ground truth could compare policy consequences conditionally; realized industrial benefit additionally requires empirical observation and intervention records.

6.3. Scope and limitations

The analysis provides a completed conditional assessment, not a fitted exposure–condition relationship, calibrated sensitivity model, or station performance estimate. The two-door model assumes a feasible response, a common loss scale, and a feasible reference schedule. Its compact form excludes false positives and requires cross-door effects to be negligible or included in the conditional consequences. A physical-failure policy needs an event and transition model. Unequal sensitivity is an illustrative input; the example does not establish an engineering mechanism or empirical range for that difference.

The available evidence may leave several models compatible with the same records. In the example, passenger records alone supply no observations of the targeted condition or inspection sensitivity, so they cannot estimate q_i and d_i without additional relationships or outcomes. Even observations of the positive-finding rate alone identify only the product $q_i d_i$ under the zero-false-positive simplification; separating prevalence from detection requires additional information. The procedure therefore retains competing assumptions and bounds its preference to the declared model set. An omitted interpretation can invalidate a preference that was stable within that set.

Application also requires reconsidering the acquisition constraint. Better condition information may be worthwhile when it changes an important action enough to justify its lifecycle burden. Rerouting, refurbishment, altered logging, or selective inspection can

change which interpretation of existing records is credible. An operational implementation needs a review trigger and a record of such changes. The conceptual classification itself neither establishes that a new monitoring system is economical nor justifies relaxing safety, accessibility, or service obligations.

7. Conclusions

Limited-Data Predictive Maintenance characterizes consequential maintenance choices for which feasible condition information remains weak and indirect evidence materially informs the assessment of future needs. Its applicability depends on the choice, horizon, and evidence, so neither a small dataset nor a particular asset class determines membership. The title's asymmetry is the mismatch between the consequences of a choice and the information available to support it.

The checklist, five-step procedure, and completed station record connect that classification to an action rationale. An exposure order becomes an inspection preference only through assumptions about condition, sensitivity, response value, and burden. Dominance filtering prevents model disagreement from reinstating a schedule that every retained model ranks below an available inspection. Subsequent choice still requires an objective: minimax regret limits opportunity loss, while maximin incremental gain protects improvement over the schedule. Their different selections identify a decision-maker commitment that additional arithmetic alone cannot settle.

The hypothetical calculation establishes these conditional implications and the sensitivity of the choice to its declared uncertainty set. Independent validation of the proxy relationships and evaluation of complete policies and assessment records remain necessary to determine operational usefulness in a particular setting.

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Supplementary Materials: Supplement S1, `two-door-calculation.zip`, contains the deterministic Python calculation, hypothetical input file, exact results, generated table, figure geometry, and regeneration instructions.

Data Availability Statement: No empirical station data were used. All hypothetical inputs are stated in Section 5.4; the calculation and exact outputs are supplied in Supplement S1.

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Abbreviations

PdM	Predictive maintenance
LD-PdM	Limited-Data Predictive Maintenance
PHM	Prognostics and health management
RUL	Remaining useful life
HVAC	Heating, ventilation, and air-conditioning
POMDP	Partially observable Markov decision process

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